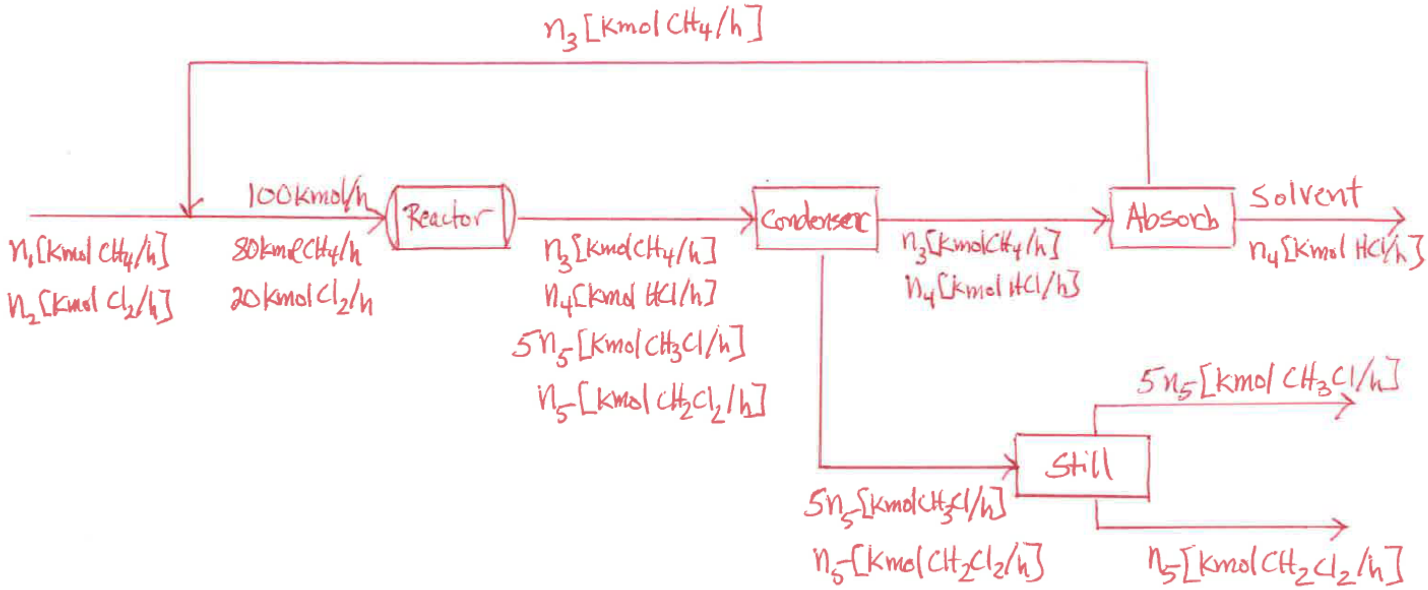
ChE 210 Homework Set #8 Due on Tuesday, 31 March 2020

(1) It is desired to produce methyl chloride (CH_3Cl) by reacting methane (CH_4) and chlorine (Cl_2). Hydrogen chloride (HCl) is another product of the reaction. The methyl chloride may react further with chlorine to produce methylene chloride (CH_2Cl_2). The process diagram is shown below. The reactor converts all of the chlorine in a single pass. The mole ratio of methyl chloride to methylene chloride in the reactor product is 5:1. The separators all achieve perfect separations as indicated in the diagram. Methane is recycled from the absorber back to the reactor feed stream.

(a) Use a basis of 100 kmol/h entering the reactor. Determine the DOF for the overall system and each single unit including the mixing point. Write the equation you would use to solve for the molar flow rate and stream compositions in the order you would solve them.

(b) Calculate the quantities in part (a) either by substitution or using a numerical method.

(c) What molar flow rates and compositions of the fresh feed and recycle streams are needed to produce 1000 kg/h of methyl chloride?



(2) Murphy 5.32

(3) Murphy 5.34 (Table B.11 in Murphy has equilibrium data for ethanol and water mixtures.)